Neural Mutual Information Estimation with Vector Copulas

Michael U. Gutmann

michael.gutmann@ed.ac.uk

School of Informatics, University of Edinburgh

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Yanzhi Chen, Zijing Ou, Adrian Weller, Michael U. Gutmann
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$$I(\mathbf{x}; \mathbf{y}) = \mathrm{KL}\left[p(\mathbf{x}, \mathbf{y}) \| p(\mathbf{x})p(\mathbf{y})\right] = \mathbb{E}_{\mathbf{x},\mathbf{y}}\left[\log \frac{p(\mathbf{x}, \mathbf{y})}{p(\mathbf{x})p(\mathbf{y})}\right] \quad (1)$$

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- Consider the case of continuous random variables.

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Previous work on estimating the MI

There are broadly two existing approaches to estimating the MI

1. Density estimation: Estimate the joint $p(\mathbf{x}, \mathbf{y})$ and the two marginals $p(\mathbf{x})$ and $p(\mathbf{y})$ separately.

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- Ratio estimation becomes harder when the MI is high (density-chasm problem, Rhodes et al, 2020)

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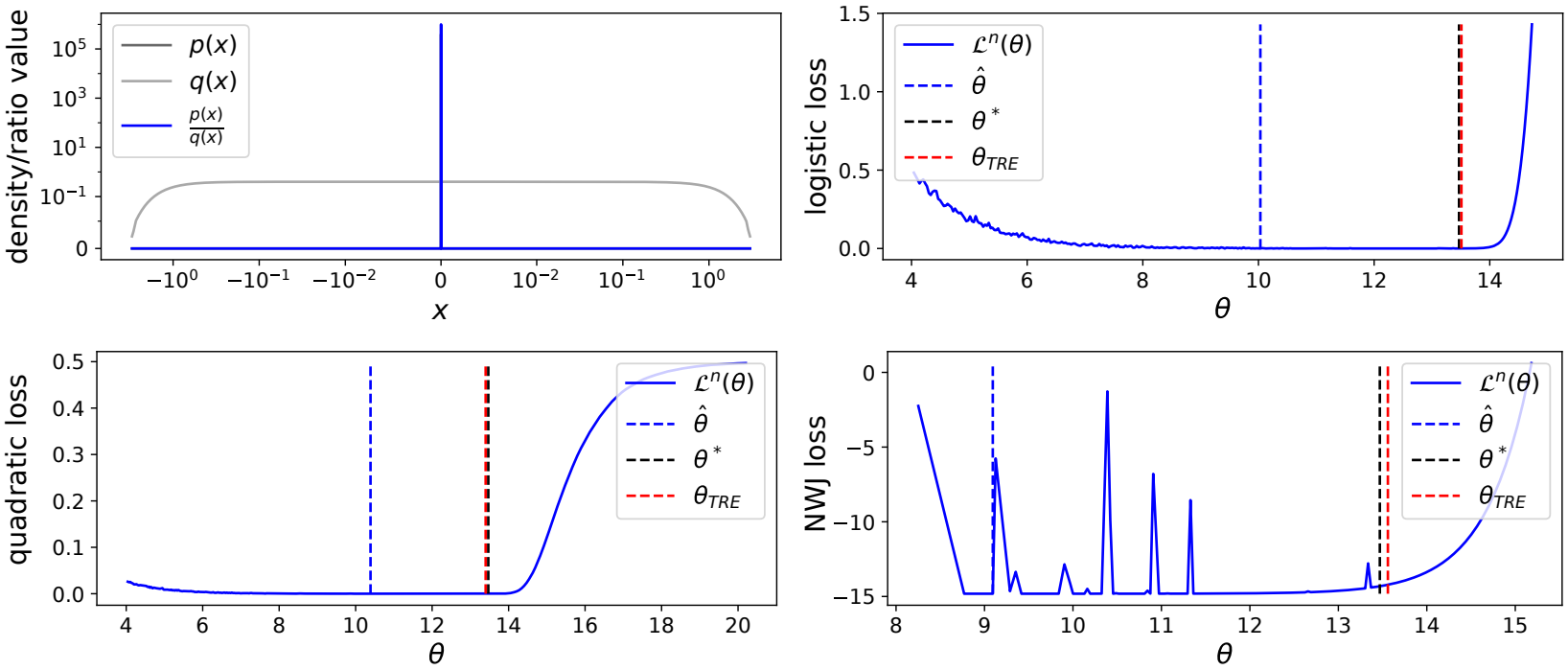
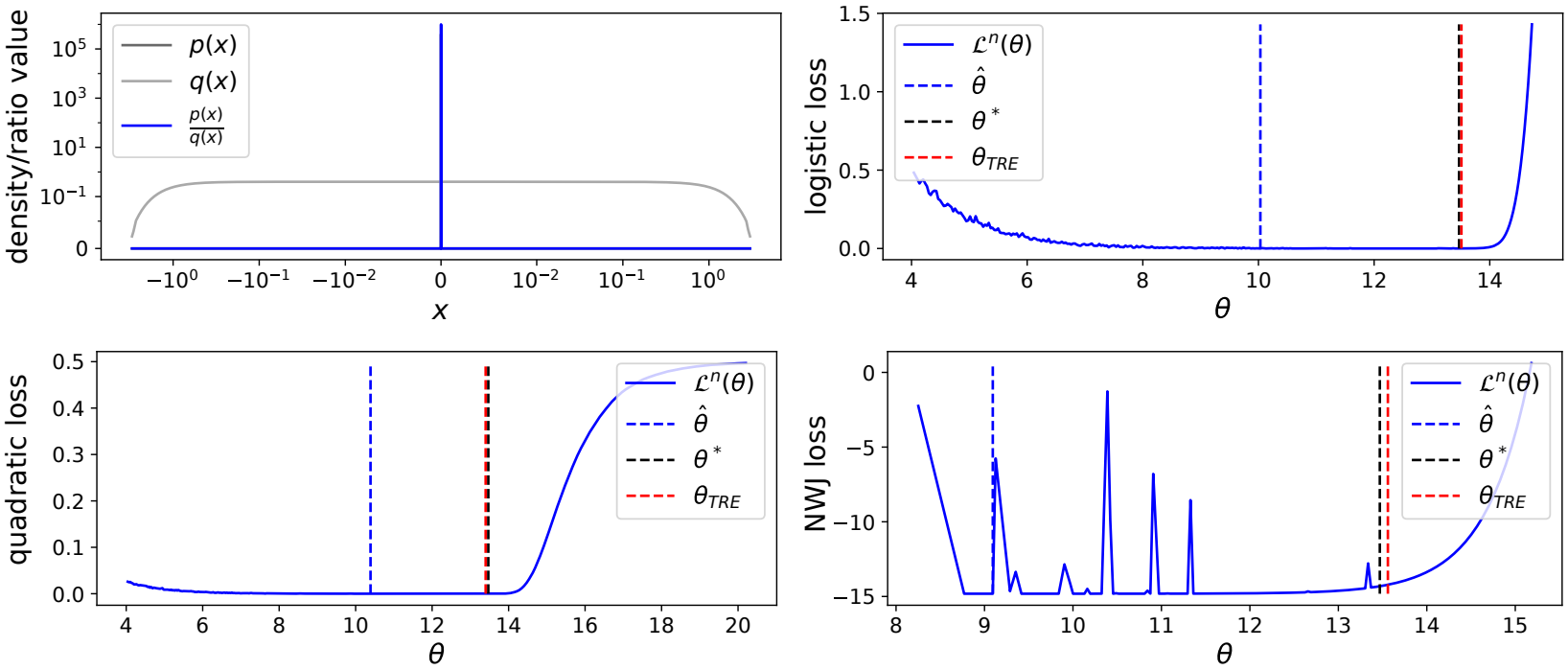


Density-chasm problem

- Logistic loss and other ratio estimation methods struggle if the two distributions are very different ("density chasm")
- Consider ratio between two zero-mean Gaussians. 10'000 samples from each distribution. Ratio parameterised by $\theta \in \mathbb{R}$.
- Solution in red bridges the "gap" using telescopic ratio estimation (TRE) (Rhodes et al, 2020)



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- We are here proposing a compromise: impose lightweight assumptions on the interactions between $\mathbf{x}$ and $\mathbf{y}$, but not on the entire distribution $p(\mathbf{x}, \mathbf{y})$.
- We model the marginals $p(\mathbf{x})$ and $p(\mathbf{y})$ flexibly but the interactions between $\mathbf{x}$ and $\mathbf{y}$ in a more controlled way, using the theory of vector copulas (Fan and Henry, J of Econometrics, 2023).

Program

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Background on copulas

- Sklar's theorem (Sklar, 1959): for any multivariate cdf $F(x_1, \ldots, x_d)$ with univariate cdfs $F_1, \ldots, F_d$, we have

$$F(x_1, \ldots, x_d) = C(F_1(x_1), \ldots, F_d(x_d)) \tag{2}$$

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- For continuous random variables, we have

$$p(x_1, \ldots, x_d) = c(F_1(x_1), \ldots, F_d(x_d)) \prod_{i=1}^{d} p_i(x_i) \tag{3}$$

  where p and c are the densities associated with F and C.

Background on copulas

$$p(x_1, \ldots, x_d) = c(F_1(x_1), \ldots, F_d(x_d)) \prod_{i=1}^{d} p_i(x_i)$$

- Variables x_i are separately transformed to uniforms $u_i = F_i(x_i)$. This is a kind of "standardisation"; transforms x_i into ranks.

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- By the change of variable formula, using $F_i'(x_i) = p_i(x_i)$, their joint density is

$$p_{\mathbf{u}}(u_1, \ldots, u_d) = c(u_1, \ldots, u_d) \tag{4}$$

Captures dependencies between the u_i, and thereby the x_i.

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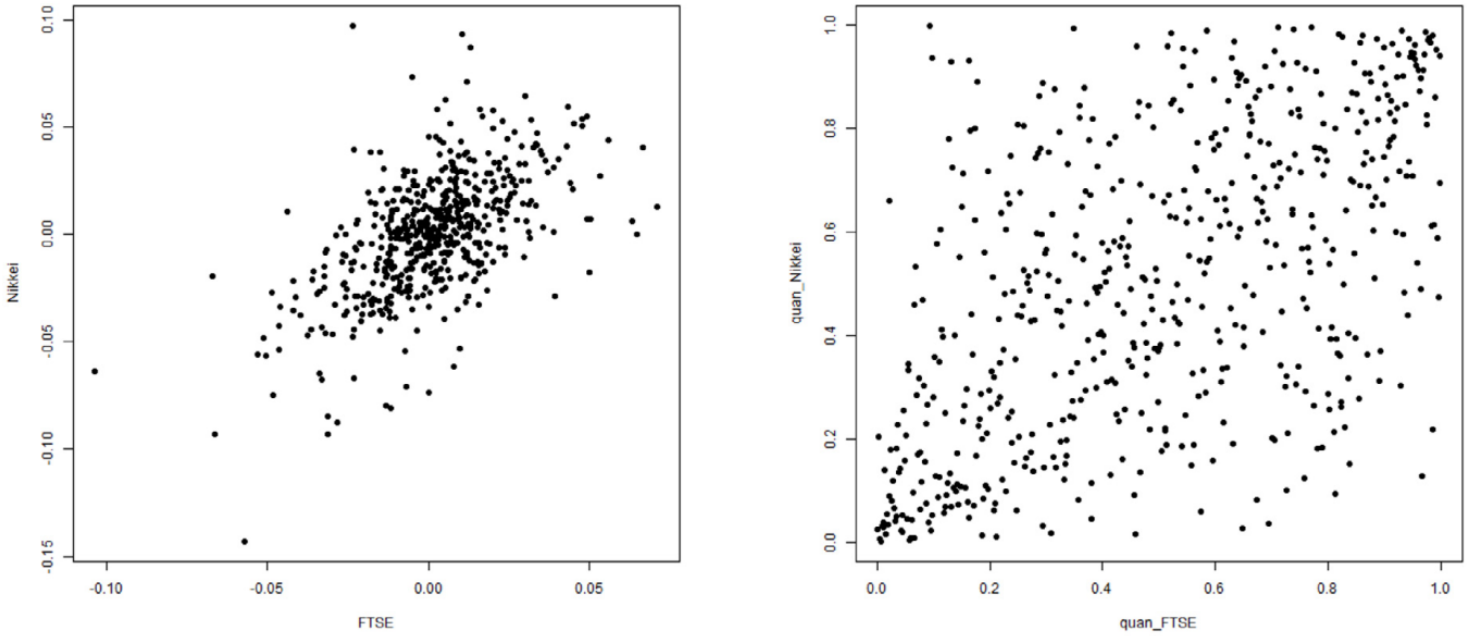
- Main point: Copula decomposition isolates dependence properties from univariate marginal properties.

Background on copulas

Example (Fan and Henry, J of Econometrics, 2023)



- Left: data in original domain (the x_i). Marginals properties overshadow finer-grained dependencies.
- Right: data in transformed domain (the u_i). Dependencies for large and small u_i (tail dependencies) become visible.

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$$G_i(\mathbf{x}_i) \sim \mathcal{U}[0,1]^{d_i} \tag{5}$$

$$p(\mathbf{x}_1, \ldots, \mathbf{x}_k) = c(G_1(\mathbf{x}_1), \ldots, G_k(\mathbf{x}_k)) \prod_{i=1}^{k} p(\mathbf{x}_i) \tag{6}$$

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- $G_i(\mathbf{x}_i)$ generalises the mappings $F(x_i)$ in the classical copulas, removing structure from $\mathbf{x}_i$ so that the vector copula only represents their dependencies, and not marginal properties.

Background on vector copulas

- Note: $\mathbf{x}_i \sim p(\mathbf{x}_i)$ and $G_i(\mathbf{x}_i) \sim \mathcal{U}[0,1]^{d_i}$ implies

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- While the $\mathbf{u}_i = G_i(\mathbf{x}_i)$ are each marginally uniformly distributed, their joint distribution is (with $\mathbf{x}_i = G_i^{-1}(\mathbf{u}_i)$)

$$\begin{aligned}
p_{\mathbf{u}}(\mathbf{u}_1, \ldots, \mathbf{u}_k) &= p(\mathbf{x}_1, \ldots, \mathbf{x}_k) \prod_{i=1}^{k} |\det J_{G_i}(\mathbf{x}_i)|^{-1} && (8)\\
&= p(\mathbf{x}_1, \ldots, \mathbf{x}_k) \prod_{i=1}^{k} \frac{1}{p(\mathbf{x}_i)} && (9)\\
&= c(G_1(\mathbf{x}_1), \ldots, G_k(\mathbf{x}_k)) \prod_{i=1}^{k} p(\mathbf{x}_i) \prod_{i=1}^{k} \frac{1}{p(\mathbf{x}_i)} && (10)\\
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- Vector copula c equals the joint distribution of the "standardised" random variables $\mathbf{u}_i$.

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Change of perspective: it's a density, not a density ratio

- Recall the definition of the MI:

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it follows from (9) on the previous slide that

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- The density ratio (pointwise MI) equals the joint density of $\mathbf{u}_x, \mathbf{u}_y$, in what follows denoted by $c(\mathbf{u}_x, \mathbf{u}_y)$.

MI as negative vector copula entropy

- With the law of the unconscious statistician, the MI thus becomes

$$I(\mathbf{x}, \mathbf{y}) = \mathbb{E}_{\mathbf{x},\mathbf{y}} \left[\log \frac{p(\mathbf{x}, \mathbf{y})}{p(\mathbf{x})p(\mathbf{y})} \right] \tag{15}$$

$$= \mathbb{E}_{\mathbf{u}_x,\mathbf{u}_y} \left[\log c(\mathbf{u}_x, \mathbf{u}_y) \right] \tag{16}$$

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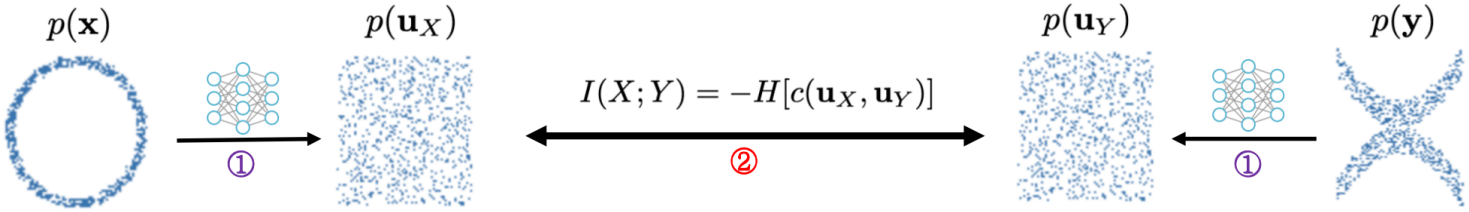
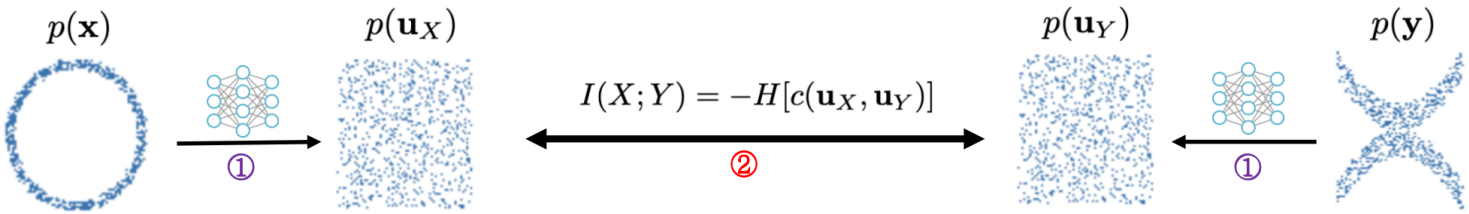
- This is the negative entropy H of c.
- Suggests a divide-and-conquer approach:



Divide and conquer approach



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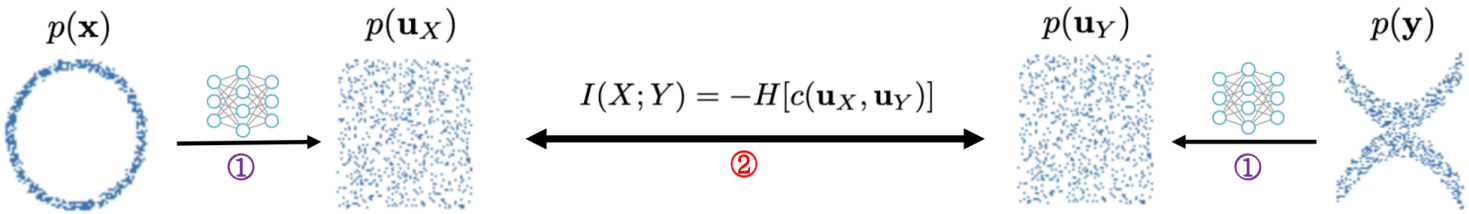

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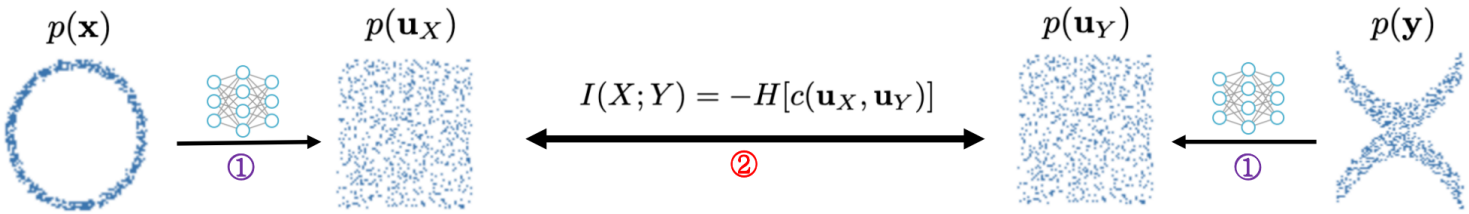

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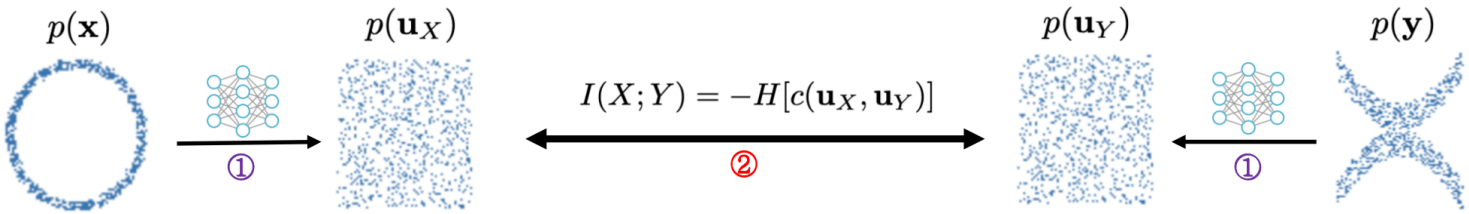

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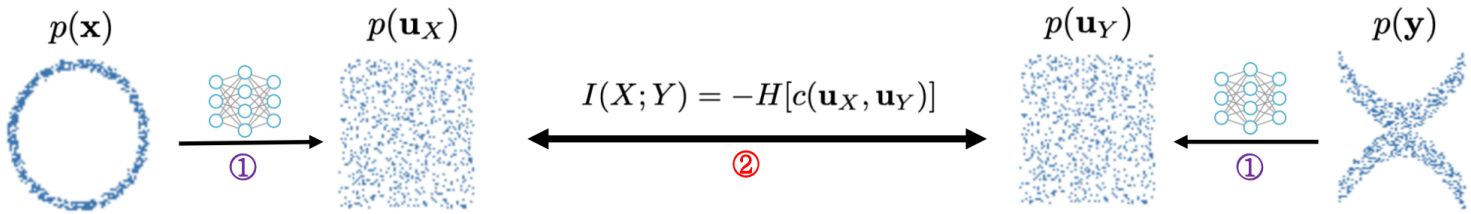

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- Note: the first step does not require joint data.

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- Here: we use Gaussian mixture models and determine the number of mixtures using validation data.

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- Parameter $\boldsymbol{\theta}$ can be learned by maximum likelihood or flow-matching (Lipman et al, 2022).

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- Our approach treats the ratio as a density in the standardised $(\mathbf{u}_x, \mathbf{u}_y)$ space, enabling standard model selection and learning by MLE, which provides the most efficient estimators.

Program

Motivation

Background

New MI estimation framework

Simulation results

Main comparison methods

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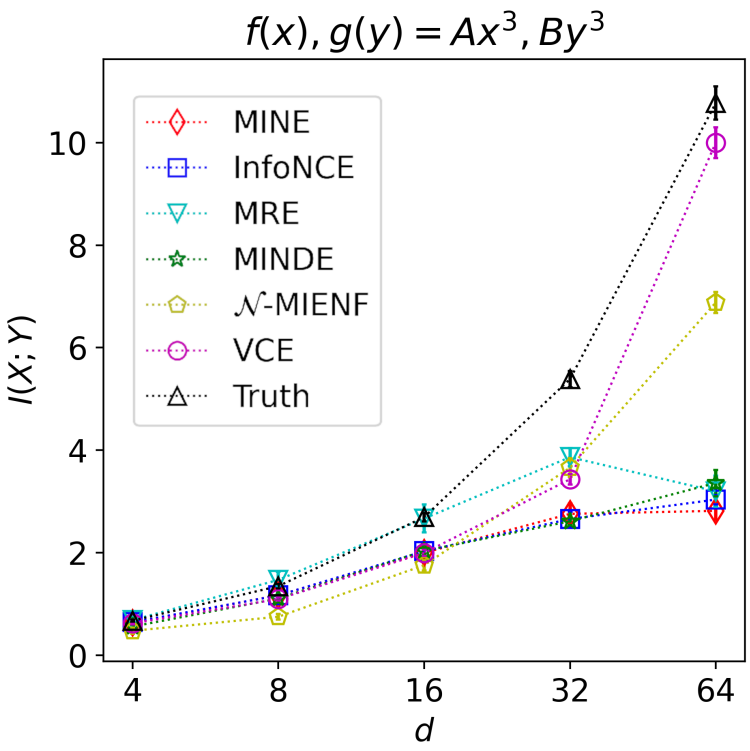
  - Multinomial ratio estimation (MRE, Srivastava et al 2023): improved version of telescopic density ratio estimation (TRE, Rhodes et al, 2020) that uses auxiliary distributions to bridge the gap between $p(\mathbf{x}, \mathbf{y})$ and $p(\mathbf{x})p(\mathbf{y})$.

Synthetic data (More results in the paper!)

- Left: Correlated Gaussian random variables x, y ($\rho = 0.7$) are nonlinearly transformed and mixed.
- Right: Student-t random variables x, y ($\rho = 0.7$) with 1 degree of freedom.
- Proposed method: Vector copula-based MI estimator (VCE)



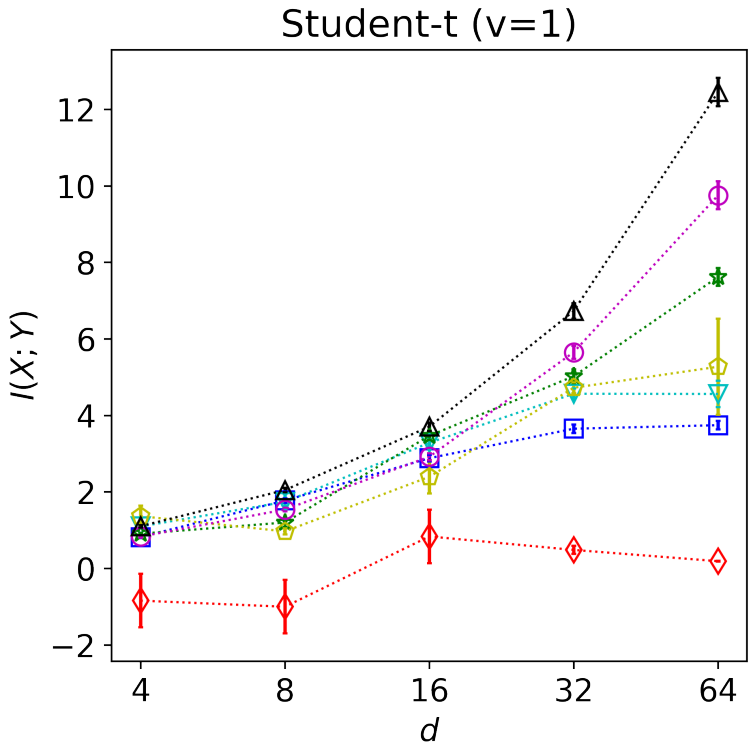
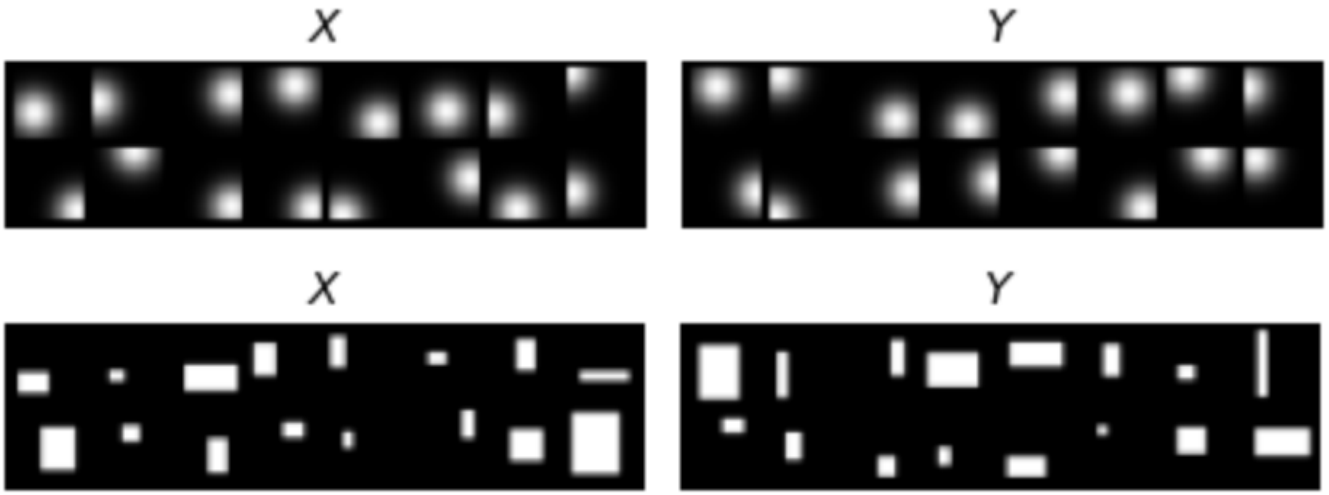


Image data (Butakov et al, 2024)

- Generated images based on random variables with known MI.
- If mapping from random variables to images is injective and piece-wise smooth, MI stays the same.
- Top: the center coordinate of the blob has been derived from a Gaussian.
- Bottom: the coordinates of the rectangles have been derived from a Gaussian.



Results on the image data

- Proposed VCE is best for blobs, still reasonable for rectangles.
- Comparison to $\mathcal{N}$-MIENF highlights the importance of using a non-Gaussian copula model (mixture model in our case)
- MRE performs well for rectangles but may overshoot (1.23 vs 1 in case of blob data)

Method	**Blobs**			**Rectangles**		
	$I(X;Y)=1$	$I(X;Y)=3$	$I(X;Y)=7$	$I(X;Y)=1$	$I(X;Y)=3$	$I(X;Y)=7$
MINE	0.89 ± 0.07	2.86 ± 0.24	5.46 ± 0.27	0.81 ± 0.13	$\mathbf{2.57 \pm 0.26}$	5.39 ± 0.23
InfoNCE	0.86 ± 0.14	2.63 ± 0.13	3.83 ± 0.12	0.78 ± 0.17	2.49 ± 0.28	3.86 ± 0.15
MRE	1.23 ± 0.16	2.85 ± 0.21	5.91 ± 0.28	0.82 ± 0.24	2.56 ± 0.48	$\mathbf{5.45 \pm 0.31}$
$\mathcal{N}$-MIENF	0.74 ± 0.12	2.42 ± 0.16	3.85 ± 0.22	0.54 ± 0.13	0.76 ± 0.14	1.54 ± 0.11
VCE	$\mathbf{0.92 \pm 0.04}$	$\mathbf{2.93 \pm 0.12}$	$\mathbf{6.53 \pm 0.36}$	$\mathbf{0.83 \pm 0.12}$	2.27 ± 0.23	5.02 ± 0.14

Results on text embeddings

- The data are pairs of embeddings computed on the Internet Movie Database (IMDB, Maas et al, 2011)
- Ground truth MI is unknown but can be estimated using the labels. MI relatively low, making task easier.
- Similar points as for images: VCE performs better than $\mathcal{N}$-MIENF, MRE may overshoot.

	X	Y
1	(positive) I thought this was a wonderful way to spend time on ...	(positive) If you like original gut wrenching laughter you will like ...
2	(negative) So im not a big fan of Boll's work but then ...	(positive) This a fantastic movie of three prisoners who become famous...

Method	$I(X;Y) \approx 2.1$	$I(X;Y) \approx 0.9$
MINE	1.83 ± 0.04	0.71 ± 0.05
InfoNCE	1.64 ± 0.09	0.70 ± 0.06
MRE	1.72 ± 0.07	1.23 ± 0.02
$\mathcal{N}$-MIENF	0.91 ± 0.05	0.43 ± 0.03
VCE	$\mathbf{2.01 \pm 0.04}$	$\mathbf{0.83 \pm 0.01}$

(a) Llama-3 13B

Method	$I(X;Y) \approx 1.5$	$I(X;Y) \approx 0.2$
MINE	$\mathbf{1.42 \pm 0.04}$	0.18 ± 0.02
InfoNCE	1.41 ± 0.03	0.19 ± 0.04
MRE	1.23 ± 0.09	0.31 ± 0.09
$\mathcal{N}$-MIENF	0.73 ± 0.03	0.11 ± 0.02
VCE	1.22 ± 0.02	$\mathbf{0.19 \pm 0.02}$

(b) BERT

Conclusions

- Estimating the MI is a fundamental and widely occurring task.
- There are two broad approaches to MI estimation:
  - density estimation
  - ratio estimation
- We here proposed a third approach based on vector copulas.
- Practical benefits:
  - Disentangles marginal from dependence structure
  - Dependence structure can be estimated via MLE
  - Flexible models for marginals, complexity control for dependence structure
- New perspective: Pointwise MI ratio can be seen as a density; MI is its (negative) entropy.
- Not always best, but provides robust and accurate MI estimates on the tested data sets.